

PRODUCT REQUIREMENTS DOCUMENT

EXECUTIVE SUMMARY

Product Name: MeetingMind

Product Vision: MeetingMind transforms meeting recordings into actionable intelligence by automatically capturing audio through the browser, generating accurate transcripts, extracting key insights, and organizing action items—all accessible through a clean web interface.

Core Purpose: Eliminates the burden of manual note-taking during meetings, allowing participants to focus on discussion while ensuring nothing important is missed. Users can revisit complete meeting records with AI-generated summaries and action items.

Target Users: Professionals who conduct frequent meetings (project managers, team leads, consultants, remote workers) and need reliable meeting documentation without dedicated note-takers.

Key MVP Features:

- Meeting Recording - User-Generated Content
- Transcript Generation - System Data
- Summary & Insights - System Data
- Action Items - User-Generated Content
- User Authentication - System/Configuration

Platform: Web application (responsive, accessible via browser on desktop/mobile devices)

Complexity Assessment: Moderate

- State Management: Backend (MongoDB) with frontend caching
- External Integrations: Open-source AI models (e.g., Whisper, Llama 3) for transcription and AI processing (reduces operational costs)
- Business Logic: Moderate (audio processing, AI orchestration, meeting lifecycle management)

MVP Success Criteria:

- Users record 30-minute meetings via browser microphone
- Transcripts generated with 90%+ accuracy
- Summaries and action items appear within 2 minutes of recording completion
- All meeting data viewable and editable through web interface
- Responsive design functions on desktop and tablet

1. USERS & PERSONAS

Primary Persona:

- Name: "Marcus the Product Manager"

- Context: Runs 3-5 meetings daily (standups, planning sessions, client calls). Struggles to participate actively while taking comprehensive notes. Often misses action items or forgets key decisions made during discussions.
- Goals: Fully engage in conversations without note-taking anxiety. Quickly review what was discussed and decided. Track action items assigned during meetings. Share meeting summaries with absent team members.
- Pain Points: Manual note-taking distracts from conversation. Post-meeting memory gaps about decisions. Action items get lost in scattered notes. No searchable meeting history.

2. FUNCTIONAL REQUIREMENTS

2.1 Core MVP Features (Priority 0)

FR-001: Meeting Recording

- Description: Browser-based audio capture from local microphone with real-time recording controls
- Entity Type: User-Generated Content
- Operations: Create (start recording), View (playback), Edit (rename, add notes), Delete (archive), List/Search (browse all recordings)
- Key Rules: Maximum 30-minute duration per recording. Browser microphone permission required. Audio stored as compressed format.
- Acceptance: Users start recording via browser, see live duration counter, stop recording, and access saved audio file

FR-002: Automatic Transcript Generation

- Description: AI-powered speech-to-text conversion creating timestamped, speaker-labeled transcripts
- Entity Type: System Data
- Operations: View (read transcript), Export (download as text/PDF), Search (find keywords)
- Key Rules: Generated automatically after recording completes. Includes timestamps every 30 seconds. Read-only after generation.
- Acceptance: Users view accurate transcript with timestamps within 90 seconds of recording completion

FR-003: AI-Generated Summary & Insights

- Description: Automated extraction of key discussion points, decisions made, and important topics covered
- Entity Type: System Data
- Operations: View (read summary), Edit (refine AI output), Export (download)
- Key Rules: Generated alongside transcript. Highlights top 5-7 key points. Editable to correct AI misinterpretations.
- Acceptance: Users see concise summary capturing main discussion themes and decisions

FR-004: Action Items Extraction

- Description: AI identifies tasks, assignments, and follow-ups mentioned during meeting
- Entity Type: User-Generated Content
- Operations: Create (AI-generated + manual add), View (list all), Edit (modify details), Delete (remove/complete), List/Search (filter by status)

- Key Rules: Each action item includes description, assignee (if mentioned), and due date (if mentioned). Users can mark complete or edit details.
- Acceptance: Users see extracted action items, manually add new ones, edit AI-generated items, and mark as complete

FR-005: User Authentication

- Description: Secure registration, login, and session management for personalized meeting access
- Entity Type: System/Configuration
- Operations: Register, Login, View profile, Edit profile, Reset password, Logout
- Key Rules: Email-based authentication. Secure password storage. Session persistence across browser sessions.
- Acceptance: Users register with email/password, login securely, and access only their own meetings

3. USER WORKFLOWS

3.1 Primary Workflow: Record and Review Meeting

Trigger: User starts a meeting and wants to capture it Outcome: User has complete meeting record with transcript, summary, and action items

Steps:

- User clicks "New Recording" and grants browser microphone permission
- System starts recording, displays live timer and waveform visualization
- User conducts meeting (up to 30 minutes), then clicks "Stop Recording"
- System saves audio locally, processes it using open-source AI models for transcription and analysis
- User sees processing status, then views complete meeting page with transcript, summary, and action items
- User reviews AI-generated content, edits action items as needed, and marks items complete
- User exports transcript or shares summary with team members

3.2 Key Supporting Workflows

Create Recording: User navigates to dashboard !' clicks "New Recording" !' g recording starts with live timer

View Meeting: User selects meeting from list !' sees full page with audio pla action items tabs

Edit Action Item: User opens meeting !' clicks action item !' modifies descrip sees updated item

Search Meetings: User enters keyword in search bar !' sees matching meetin content, or date

Export Transcript: User opens meeting !' clicks "Export" !' selects format (TX

4. BUSINESS RULES

4.1 Entity Lifecycle Rules

Meeting Recording:

- Type: User-Generated Content
- Creation: All authenticated users
- Editing: Owner only (rename, add manual notes)
- Deletion: Owner only - soft delete (archived for 30 days, then permanently removed)

Transcript:

- Type: System Data
- Creation: Auto-generated by system after recording completes
- Editing: Not allowed - immutable record (users can export and edit externally)
- Deletion: Deleted when parent meeting is permanently deleted

Summary & Insights:

- Type: System Data
- Creation: Auto-generated by system after recording completes
- Editing: Owner can edit to refine AI output (preserves original version)
- Deletion: Deleted when parent meeting is permanently deleted

Action Item:

- Type: User-Generated Content
- Creation: Auto-generated by AI + manual creation by owner
- Editing: Owner only (modify description, assignee, due date, status)
- Deletion: Owner only - soft delete (marked complete or archived)

User:

- Type: System/Configuration
- Creation: Self-registration via signup form
- Editing: Owner only (profile details, password)
- Deletion: Owner can delete account (all meetings permanently removed after 30-day grace period)

4.2 Data Validation Rules

Meeting Recording:

- Required Fields: title, audioFile, ownerId, recordedAt, duration
- Field Constraints: Title minimum 3 characters, maximum 100 characters; duration maximum 30 minutes (1800 seconds); audio file must be valid format (WebM/MP3)

Transcript:

- Required Fields: meetingId, content, generatedAt
- Field Constraints: Content must be non-empty string; timestamps in format "MM:SS"

Summary & Insights:

- Required Fields: meetingId, summary, keyPoints, generatedAt
- Field Constraints: Summary minimum 50 characters; keyPoints array with 3-7 items

Action Item:

- Required Fields: meetingId, description, status
- Field Constraints: Description minimum 5 characters, maximum 500 characters; status must be one of: pending, in-progress, completed; due date must be present or future date if provided

User:

- Required Fields: email, password, name
- Field Constraints: Valid email format; password minimum 8 characters with letter and number; name minimum 2 characters

4.3 Access & Process Rules

- Users can only view/edit their own meetings and action items
- Recording automatically stops at 30-minute mark
- Transcript and summary generation begins immediately after recording stops
- Processing typically completes within 90-120 seconds for 30-minute recordings
- Archived meetings retained for 30 days before permanent deletion
- Maximum 100 meetings per user in free tier
- Audio files compressed to optimize storage (target: ~1MB per minute)

5. DATA REQUIREMENTS

5.1 Core Entities

User

- Type: System/Configuration | Storage: Backend (MongoDB)
- Key Fields: id, email, passwordHash, name, createdAt, preferences, storageUsed
- Relationships: has many Meetings, has many ActionItems
- Lifecycle: Full CRUD with export and account deletion (30-day grace period)

Meeting

- Type: User-Generated Content | Storage: Backend (MongoDB + local file system for audio)
- Key Fields: id, title, audioFilePath, duration, recordedAt, ownerId, status (recording/processing/completed/archived), createdAt, updatedAt
- Relationships: belongs to User, has one Transcript, has one Summary, has many ActionItems

- Lifecycle: Create, View, Edit (title/notes), Delete (soft delete with 30-day retention), List/Search

Transcript

- Type: System Data | Storage: Backend (MongoDB)
- Key Fields: id, meetingId, content (full text), timestampedSegments (array), generatedAt, processingTime
- Relationships: belongs to Meeting
- Lifecycle: View, Export only (immutable after generation)

Summary

- Type: System Data | Storage: Backend (MongoDB)
- Key Fields: id, meetingId, summaryText, keyPoints (array), decisions (array), generatedAt, editedAt, originalVersion
- Relationships: belongs to Meeting
- Lifecycle: View, Edit (preserves original), Export

ActionItem

- Type: User-Generated Content | Storage: Backend (MongoDB)
- Key Fields: id, meetingId, description, assignee, dueDate, status, source (ai-generated/manual), createdAt, updatedAt, completedAt
- Relationships: belongs to Meeting, belongs to User
- Lifecycle: Full CRUD + mark complete + archive

5.2 Data Storage Strategy

- Primary Storage: MongoDB for all structured data (users, meetings metadata, transcripts, summaries, action items)
- Audio Storage: Local server file system (for MVP simplicity and cost reduction) with file paths referenced in MongoDB
- Capacity: Estimated 1MB per minute of audio; 30-minute meeting = ~30MB audio + ~50KB text data. Local server must be provisioned with adequate disk space.
- Persistence: All data persists indefinitely until user deletion or 30-day archive expiration
- Audit Fields: All entities include createdAt, updatedAt, createdBy (userId)

6. INTEGRATION REQUIREMENTS

Open-Source Speech-to-Text (e.g., faster-whisper or Whisper via Groq/Together AI):

- Purpose: Speech-to-text transcription with speaker diarization and timestamps
- Type: Internal microservice (self-hosted) or cost-effective open-source API provider
- Data Exchange: Sends audio file (WebM/MP3), Receives JSON with timestamped transcript segments
- Trigger: Automatically called when recording status changes to "processing"
- Error Handling: Retry up to 3 times; if fails, mark meeting as "transcription-failed" and notify user
- Cost Optimization: Utilizing open-source Whisper weights eliminates proprietary API costs. Can be hosted on dedicated GPU instances or utilized via serverless open-source API providers.

Open-Source LLM (e.g., Llama 3 8B/70B or Mixtral 8x7B):

- Purpose: Generate meeting summary, extract key points, identify action items
- Type: Internal microservice (e.g., vLLM, Ollama) or open-weight API provider
- Data Exchange: Sends transcript text with prompt instructions, Receives JSON with summary, keyPoints array, actionItems array
- Trigger: Called immediately after successful transcription
- Error Handling: Retry up to 3 times; if fails, provide transcript only and allow manual summary creation
- Cost Optimization: Replaces expensive GPT-4 calls with highly capable open-source models, drastically reducing the per-meeting processing costs while maintaining high-quality extraction.

7. VIEWS & NAVIGATION

7.1 Primary Views

Dashboard (/) - Recent meetings list with status indicators, quick "New Recording" button, storage usage meter, upcoming action items summary

Recording View (/record) - Live recording interface with waveform visualization, timer, pause/resume/stop controls, microphone level indicator

Meeting Detail (/meetings/:id) - Tabbed interface: Overview (audio player, metadata), Transcript (timestamped text), Summary (key points, decisions), Action Items (list with status)

Meetings List (/meetings) - All meetings with search/filter (date range, status), sort options, grid/list view toggle, bulk actions (archive, export)

Action Items (/actions) - Consolidated view of all action items across meetings, filterable by status/due date/meeting, sortable, quick complete toggle

Settings (/settings) - Profile management, password change, storage usage details, export all data, delete account

7.2 Navigation Structure

Main Nav: Dashboard | Meetings | Action Items | Settings | User Menu (profile, logout) Default Landing: Dashboard Mobile: Bottom navigation bar with icons, hamburger menu for secondary options, responsive recording interface

8. MVP SCOPE & CONSTRAINTS

8.1 MVP Success Definition

The MVP is successful when:

- ' Users record 30-minute meetings via browser microphone without technical issues
- ' Transcripts generated with 90%+ accuracy within 2 minutes
- ' Summaries capture main discussion points accurately
- ' Action items extracted and editable by users
- ' All CRUD operations work for meetings and action items

- ' Responsive design functions on desktop and tablet browsers
- ' Data persists in MongoDB and accessible across sessions
- ' Audio playback works reliably with seek controls

8.2 In Scope for MVP

Core features included:

- FR-001: Meeting Recording (browser-based audio capture)
- FR-002: Automatic Transcript Generation (Open-Source Whisper)
- FR-003: AI-Generated Summary & Insights (Open-Source LLM, e.g., Llama 3)
- FR-004: Action Items Extraction (AI + manual)
- FR-005: User Authentication (email/password)
- Audio playback with seek controls
- Search meetings by title and content
- Export transcripts and summaries
- Edit AI-generated summaries and action items
- Archive/delete meetings with 30-day retention

8.3 Technical Constraints

- Data Storage: MongoDB for structured data, local server file system for audio files
- Concurrent Users: Expected 100-500 users during MVP phase
- Performance: Recording starts <1s after permission granted; transcript generation <2 minutes for 30-minute meeting; page loads <2s
- Browser Support: Chrome, Firefox, Safari, Edge (last 2 versions) - requires MediaRecorder API support
- Mobile: Responsive design for tablet browsers; phone browsers supported for viewing only (recording requires desktop/tablet)
- Offline: Not supported - requires internet connection for recording and AI processing
- Audio Quality: Minimum 16kHz sample rate for accurate transcription
- Recording Limit: Hard stop at 30 minutes to manage processing costs and storage

8.4 Known Limitations

For MVP:

- Browser microphone only (no phone dial-in or external audio sources)
- Single speaker optimization (multi-speaker diarization may have lower accuracy)
- English language only for transcription and AI processing
- No real-time transcription during recording (post-processing only)
- No collaborative editing (single owner per meeting)
- No calendar integration or scheduled recordings
- No video capture (audio only)
- Storage limited to 100 meetings per user

Future Enhancements:

- V2 will add real-time transcription, multi-language support, calendar integration, team collaboration features, and video recording capabilities

9. ASSUMPTIONS & DECISIONS

9.1 Platform Decisions

- Type: Full-stack web application (React frontend + FastAPI (Python) backend)
- Storage: MongoDB for structured data + local server file system for audio files (MVP)
- Auth: Backend JWT-based authentication with secure password hashing
- AI Processing: Server-side AI processing (FastAPI integrates seamlessly with Python-based AI ecosystems like vLLM, Ollama, or direct model execution)

9.2 Entity Lifecycle Decisions

Meeting Recording: Full CRUD + soft delete with 30-day retention

- Reason: User-generated content requiring full control; soft delete allows recovery from accidental deletion

Transcript: View + Export only (immutable)

- Reason: Maintains integrity of original AI-generated record; users can export and edit externally if needed

Summary & Insights: View + Edit (preserves original)

- Reason: Allows users to refine AI output while maintaining audit trail of original generation

Action Items: Full CRUD + status management

- Reason: User-generated content (even if AI-seeded) requiring full lifecycle management for task tracking

9.3 Key Assumptions

- Users have reliable microphone hardware and quiet recording environments - Reasoning: Product targets professionals conducting planned meetings, not casual/impromptu conversations in noisy settings
- 30-minute limit sufficient for majority of target meetings - Reasoning: Standups, planning sessions, and client calls typically under 30 minutes; longer meetings can be split into segments
- Post-processing acceptable (no real-time transcription required for MVP) - Reasoning: Reduces technical complexity significantly; users can wait 1-2 minutes after meeting for results
- Single-user ownership model sufficient for MVP - Reasoning: Simplifies access control and data model; team collaboration deferred to V2
- Open-source AI costs manageable at MVP scale - Reasoning: Utilizing self-hosted or open-weight API providers (Whisper, Llama 3) drastically reduces per-meeting processing costs compared to proprietary APIs. FastAPI backend allows efficient handling of these Python-native AI workloads.
- Local server storage is sufficient for MVP audio files - Reasoning: Storing compressed audio locally avoids cloud storage setup and costs during the initial MVP validation phase. Server disk space will be monitored and provisioned accordingly.

9.4 Clarification Q&A Summary

Q: How should the recordings and generated notes be stored? A: In MongoDB Decision: Backend MongoDB for all structured data (meetings, transcripts, summaries, action items) with local file system paths for audio files

Q: Will users need the ability to manually edit the generated transcripts, summaries, and action items if the AI makes a mistake? A: Yes Decision: Transcripts remain immutable (export to edit externally); summaries editable with original version preserved; action items fully editable as user-generated content

Q: Once the meeting notes and action items are generated, how will users use them? A: Revisit them after the meeting ends Decision: Optimized for post-meeting review workflow with dedicated meeting detail view, searchable archive, and action items dashboard

Q: Are there expected limits on the meeting duration? A: Up to 30 minutes Decision: Hard limit at 30 minutes with automatic stop; optimizes processing time, storage costs, and aligns with typical meeting lengths